

Ahmed Seddik

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PROFESSIONAL PROFILE

Data Engineer and Python developer with production experience building ETL/ELT pipelines, multi-source data processing systems, and ML-integrated analytics platforms. Engineered pipelines handling 1M+ log entries with measurable latency and accuracy improvements at an industrial analytics company; delivered a live SaaS platform processing real-time inspection data for a paying client in the field. Proficient in PostgreSQL, FastAPI, Docker, and AWS data infrastructure. Available for remote or hybrid roles; open to relocation.

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines, Batch Processing, Data Modelling, Feature Engineering, Anomaly Detection, Workflow Orchestration, Vector Databases (ChromaDB)

Data & ML Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, LightGBM, XGBoost, Prophet, LangChain, spaCy, OpenCV, Plotly, Hugging Face

Languages: Python, SQL, TypeScript, JavaScript, Java, C#

Backend & APIs: FastAPI, SQLAlchemy, Alembic, Celery, Node.js / Express, REST APIs, .NET / WPF

Databases & Storage: PostgreSQL, SQLite, Redis, AWS S3, ChromaDB

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, VPC, IAM, CloudWatch), Docker, GitHub Actions CI/CD, Nginx, Linux

Tools: Jupyter, Git, Postman, VSCode, Docker Compose

EXPERIENCE

Data Engineer Intern Jan – Jul 2025

ACCUR DIGITAL, Remote

- Engineered an end-to-end image data pipeline — ingesting 5,000+ raw industrial corrosion images, applying preprocessing transforms and augmentation, training a U-Net + ResNet34 segmentation model, and writing structured severity classifications to a downstream datastore; achieved 90% validation accuracy.
- Built Python ETL pipelines ingesting and transforming 1M+ industrial log entries across multiple source systems; optimised preprocessing and feature extraction stages, reducing downstream anomaly detection latency by 20%.
- Designed a text processing pipeline over 10,000+ unstructured maintenance records — TF-IDF vectorisation, unsupervised clustering, and structured feature extraction — lifting scheduling-prediction model accuracy by 25%.
- Built a batch model scoring pipeline using LightGBM/XGBoost failure-prediction models ($R^2 > 0.85$), integrating outputs into a live industrial analytics dashboard with scheduled retraining cycles.

Cloud Architect Intern Aug – Sep 2025

National Telecommunication Institute (NTI), Remote

- Designed and secured scalable AWS environments using EC2, RDS, S3, Lambda, IAM, VPC, and CloudWatch, applying high-availability and cost-optimisation best practices.
- Completed certification-track training aligned with AWS Solutions Architect Associate competencies.

SELECTED PROJECTS

NDT Inspection Management Platform — MASSA Consulting Ltd. & SA International — React · TypeScript · FastAPI · PostgreSQL · Docker · C# · WPF · .NET

- Delivered as a freelance paid engagement for two industrial clients (MASSA Consulting Ltd. & SA International) — a full SaaS web platform paired with a WPF desktop app automating the end-to-end NDT inspection process: job scheduling, multi-method workflow execution, equipment and certificate tracking, and automated PDF report generation; reduced human error by 60% and accelerated the overall process by 80%.
- JWT/RBAC auth with five role-separated dashboards (Admin, Manager, Reviewer, Inspector, Client); Celery async pipeline for report rendering; WPF companion app with local-first data capture and sync to the central PostgreSQL database.

Corrosion Annotator — Hybrid CV Segmentation System — TensorFlow · U-Net · ResNet34 · OpenCV · CRF

- Designed a reproducible batch-processing pipeline for industrial corrosion imagery: multi-source image ingestion, U-Net + ResNet34 deep learning segmentation combined with HSV classical vision, and CRF post-processing refinement — outputting structured severity labels, area ratios, and annotated overlays per image.
- Modular pipeline architecture separates ingestion, inference, and output stages; applicable to pipeline, storage tank, and offshore infrastructure inspection datasets.

Email Security Backend — NLP Threat Detection API — FastAPI · PostgreSQL · TF-IDF / LinearSVC · VirusTotal · Safe Browsing

- Built a multi-signal data ingestion and scoring pipeline: SPF/DKIM/DMARC header parsing, URL reputation lookups (VirusTotal / Safe Browsing API), attachment hash analysis, and TF-IDF/LinearSVC phishing classification — funnelled into a unified threat score per message with auto-quarantine or delete actions.
- Prometheus metrics endpoint for pipeline monitoring; token-bucket rate limiter managing external API quotas; deployed on Render/Railway with PostgreSQL persistence.

Business Wise — AI Maintenance Analytics — Python · XGBoost · Prophet · LangChain · OpenAI · spaCy

- Built a multi-stage analytics pipeline processing industrial equipment telemetry: Isolation Forest and LOF anomaly detection on sensor streams, XGBoost failure-prediction batch scoring, and Prophet time-series forecasting — all orchestrated into a single end-to-end batch run from raw data to structured output.
- Built a spaCy NLP extraction layer over unstructured maintenance work-order text; extracted entities are structured into features and fed into the forecasting pipeline, replacing manual data-entry bottlenecks.
- Generated LLM-powered executive summaries (LangChain + OpenAI) from pipeline outputs; delivered results as interactive Plotly dashboards and auto-generated PDF reports.

EDUCATION

B.Eng., Computer Engineering

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Alexandria, Egypt

Relevant Coursework: Data Analytics · Data Security & Cryptography · Advanced Computer Networks · Database Systems · Data Structures & Algorithms · OOP · System Programming · Operations Research · Micro-Processor · Operating System

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization — National Telecommunication Institute (NTI), 2024
- AWS Cloud Architecture Training (Solutions Architect Associate track) — NTI, 2025
- ECPC (Egyptian Collegiate Programming Contest) Participant, 2024
- National Champion, RoboCup Junior Egypt, 2019 · Asia-Pacific RoboCup Participant, 2018